

Introduction to Artificial Intelligence

Artificial Intelligence (AI) is a branch of computer science that focuses on creating systems capable of performing tasks that normally require human intelligence. These tasks include learning, reasoning, problem-solving, perception, and language understanding.

AI can be classified into three categories:

1. Narrow AI – designed for specific tasks such as virtual assistants and recommendation systems.
2. General AI – human-like intelligence capable of performing a wide range of tasks.
3. Super AI – a hypothetical intelligence surpassing human capabilities.

Machine Learning is a subset of AI that enables systems to learn from data without being explicitly programmed.

Deep Learning is a subset of Machine Learning that uses artificial neural networks with multiple layers. It is widely used in image recognition, speech recognition, and natural language processing.

Applications of AI include Healthcare, Education, Finance, Transportation, and Cybersecurity.

Advantages of AI: - Increased efficiency - Reduced human error - Automation of repetitive tasks

Challenges of AI: - Data privacy concerns - Bias in algorithms - High computational costs

Artificial Intelligence continues to transform industries and improve decision-making processes across the world.